

# DSCI445 - Homework 3

Your Name

Be sure to `set.seed(445)` at the beginning of your homework.

```
#reproducibility
set.seed(445)
```

1. In this exercise you will create some simulated data and will fit simple linear regression models to it. Make sure to use `set.seed(445)` prior to starting part (a) to ensure reproducible results.

- (a) Using the `rnorm()` function, create a vector  $x$  containing 100 observations drawn from a  $N(0, 1)$  distribution. This represents the feature,  $X$ .
- (b) Using the `rnorm()` function, create a vector `eps` containing 100 observations drawn from a  $N(0, 0.25)$  distribution, i.e. a Normal distribution with mean zero and variance 0.25.
- (c) Using  $x$  and `eps`, generate a vector  $y$  according to the model

$$Y = -1 + 0.5X + \epsilon.$$

What is the length of the vector  $y$ ? What are the values of  $\beta_0$  and  $\beta_1$  in this linear model?

- (d) Create a scatterplot displaying the relationship between  $x$  and  $y$ . Comment on what you observe.
- (e) Fit a least squares linear model to predict  $y$  from  $x$ . Comment on the model obtained. How do  $\hat{\beta}_0$  and  $\hat{\beta}_1$  compare to  $\beta_0$  and  $\beta_1$ ?
- (f) Display the least squares line on the scatterplot obtained in (d) in blue. Draw the population regression line on the plot in red. (See `geom_abline()` for how to add a line based on intercept and slope.)
- (g) Now fit a polynomial regression model that predicts  $y$  using  $x$  and  $x^2$ . Is there evidence that the quadratic term improves the model fit? Explain your answer.
- (h) Repeat (a)-(f) after modifying the data generation process in such a way that there is *less* noise in the data. The model should remain the same. You can accomplish this by changing the variance of the normal distribution used to generate the error term  $\epsilon$  in (b). Describe your results.
- (i) Repeat (a)-(f) after modifying the data generation process in such a way that there is *more* noise in the data. The model should remain the same. You can accomplish this by changing the variance of the normal distribution used to generate the error term  $\epsilon$  in (b). Describe your results.
- (j) What are the confidence intervals for  $\beta_0$  and  $\beta_1$  based on the original data set, the noisier data set, and the less noisy data sets? Comment on your results.

```
# simulated regression task
```

2. When the number of features  $p$  is large, there tends to be a deterioration in the performance of KNN and other *local* approaches that perform prediction using only observations that are *near* the test observation for which a prediction must be made. This phenomenon is known as the *curse of dimensionality*, and it ties into the fact that non-parametric approaches often perform poorly when  $p$  is large. We will now investigate this curse.

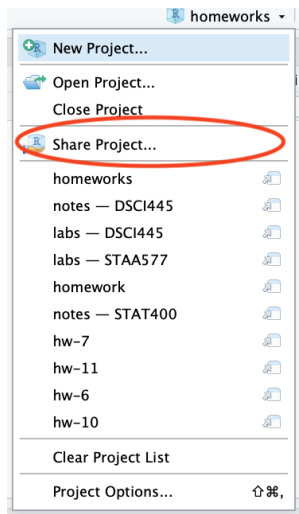
- (a) Suppose that we have a set of observations, each with measurements on  $p = 1$  feature,  $X$ . We assume that  $X$  is uniformly (evenly) distributed on  $[0, 1]$ . Associated with each observation is a response value. Suppose that we wish to predict a test observation's response using only observations that are within 10% of the range of  $X$  closest to that test observation. For instance, in order to predict the response for a test observation with  $X = 0.6$  we will use observations in the range  $[0.55, 0.65]$ . On average, what fraction of the available observations will we use to make this prediction?
- (b) Now suppose that we have a set of observations, each with measurements on  $p = 2$  features,  $X_1$  and  $X_2$ . We assume  $(X_1, X_2)$  are uniformly distributed on  $[0, 1] \times [0, 1]$ . we wish to predict a test observation's response using only observations that are within 10% of the range of  $X_1$  and within 10% of the range of  $X_2$  closest to that test observation. For instance, in order to predict the response for a test observation with  $X_1 = 0.6$  and  $X_2 = 0.35$  we will use observations in the range  $[0.55, 0.65]$  for  $X_1$  and in the range  $[0.3, 0.4]$  for  $X_2$ . On average, what fraction of the available observations will we use to make this prediction?
- (c) Now suppose that we have a set of observations, each with measurements on  $p = 100$  features. Again, the observations are uniformly distributed on each feature, and again each feature ranges from 0 to 1. we wish to predict a test observation's response using only observations that are within 10% of each feature's range that is closest to that test observation. What fraction of the available observations will we use to make this prediction?
- (d) Using your answers to (a)–(c), argue that a drawback of KNN when  $p$  is large is that there are very few training observations *near* any given test observation.
- (e) Now suppose that we wish to make a prediction for a test observation by creating a  $p$ -dimensional hypercube centered around the test observation that contains, on average, 10% of the training observations. For  $p = 1, 2$ , and 100, what is the length of each side of the hypercube? Comment on your answer.

[Hint: A hypercube is a generalization of a cube to an arbitrary number of dimensions. When  $p = 1$ , a hypercube is simple a line segment. When  $p = 2$  it is a square, and when  $p = 100$  it is a 100-dimensional cube.]

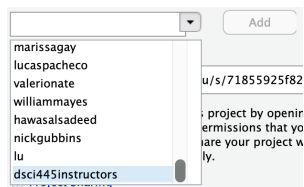
Turn in in a pdf of your homework to canvas using the provided Rmd file as a template. Your Rmd file on the server will also be used in grading, so be sure they are identical.

**Be sure to share your server project with the instructor and grader. You only need to do this once per semester.**

1. Open your **homeworks** project on [liberator.stat.colostate.edu](https://liberator.stat.colostate.edu)
2. Click the drop down on the project (top right side) > Share Project...



3. Click the drop down and add “dsci445instructors” to your project.



This is how you **receive points** for reproducibility on your homework!